Itgan at AraSeg Shared Task 2026: Diversify, Don't Curate – Learned Ensemble Combination for Arabic Sentence Segmentation

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Abstract

We describe our submissions to the four subtasks of the AraSeg 2026 shared task in both tracks. Our components are conventional word-level boundary taggers, and our contribution is a measurement of what combining them is worth. Averaging 21 members gains $+0.89$ F1 over the best single model, and replacing the uniform probability average with a learned logistic combiner adds a further $+0.23$ ($+1.12$ in total), reaching significance only on the two subtasks that lack punctuation. Both are smaller than the encoder swap ($+1.82$). What the combiner buys is freedom from curation. Added to the base encoder, each of six different pretrained encoders helps under the combiner ($+0.12$ to $+0.72$, mean $+0.34$), whereas under averaging three of the six are harmful, by as much as 0.62. Another seed of an existing encoder is worth nothing under either rule; members should therefore be *diversified* rather than *curated*, provided the weights are learned. We also report three null results for inference-time structure, including a semi-Markov CRF whose failure we trace to a length prior that is uninformative at corpus level. Our closed-track systems placed fifth and seventh where ranked. In the open track, pretraining on 3.2M tokens of silver boundaries from Arabic Wikipedia before fine-tuning on the gold training split is worth $+0.58$ F1, and two of the closed-track findings reproduce on that external data.

1 Introduction

Sentence segmentation is a preprocessing step for most downstream Arabic NLP, and a hard one in a language where punctuation is inconsistent, ambiguous, and in long stretches absent. The AraSeg 2026 shared task (Elkholy et al., 2026b) formalises this as binary token classification across four conditions varying the availability of punctuation and paragraph structure. There are two tracks, *closed* (training restricted to the released `train` split) and *open* (any publicly available data). The four conditions expose the difficulty. Our F1 on the Blind Test set falls from 94.4 on documents with punctuation and paragraph structure to 86.5 on documents with neither, where every boundary must be recovered from lexical evidence alone.

Our participation began as a replication of the organisers' CAMeLBERT baseline and became an attempt to measure where performance in such systems resides. The answer is blunt. It resides in the training data and the encoder, and almost nowhere else that we could find.

The prevailing recipe for tasks of this kind is to select a strong monolingual encoder, tune it, and ensemble a handful of top-scoring variants by probability averaging, curating the members by validation score. We ablated every stage of that recipe. Choosing the right encoder is worth $+1.82$ F1, and the volume of training data matters more still (Section 2); everything downstream of those two is small. Averaging 21 members buys $+0.89$ over the best single model, and learning the combination weights rather than averaging adds $+0.23$ on top of that. Inference-time structure contributes nothing at all, with expected F1 decoding, minimum-segment-length constraints and an exact semi-Markov CRF all null (Appendix A), and synthetic augmentation of the closed-track data actively hurts, with a dose-response.

The one place the combination rule earns its keep is curation. Under uniform averaging, admitting models below the ensemble's level costs accuracy, so practitioners screen candidates by standalone score, and they are right to do so, since three of the six distinct encoders we tried damage an average by up to 0.62. Under a learned combiner all six help instead, so the screening step can simply be dropped. Standalone accuracy is a poor guide in any case, since our most heavily weighted member scores 0.6 F1 *below* the base encoder in isolation and, across the full ensemble, fitted weight barely

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tracks standalone F1. The systems built this way placed fifth and seventh on the two closed-track subtasks where our entry is ranked.

Contributions.

1. Evidence that ensemble members should be diversified rather than curated, and that this holds only when combination weights are learned. Every distinct encoder helps under a learned combiner, while half of them hurt under an average (Section 5).
2. A calibrated account of how little the combination stage is worth once its baseline is computed properly, together with a partial mechanism for where it pays. Mean pairwise member disagreement rises monotonically across the four subtasks, but this order does not exactly match combination gains at the subtask level, since one subtask (NP) has the second-lowest disagreement of the four yet the only negative gain. Within a subtask, however, documents where members disagree more are documents where the combiner gains more over probability averaging, in the two no-punctuation subtasks (Section 5).
3. Three null results for inference-time structure, including an exact semi-Markov CRF whose failure we dissect with an entropy analysis and an oracle decomposition. A document-level length prior would be worth $+3.1$ F1, about 38% of it from the sentence count alone, but none of it is attainable from corpus-level statistics (Appendix A).

2 Background

The AraSeg corpus (Elkholy et al., 2026a) is released as 658 documents in four parallel variants. PA has punctuation and paragraph breaks, NP punctuation only, NoPnx-PA paragraphs only, and NoPnx-NP neither. Each variant is split 174 train / 222 dev / 262 test, and all four share document identifiers.

Two properties shaped our approach. First, the shared task ships the same dev and test splits as Elkholy et al. (2026a) but only 10.2% of its training documents (174 versus 1,703, and 128K versus 1.1M tokens), while their hyperparameters and ours are essentially the same. The published headline is therefore not a target attainable within the closed track, and closed-track systems should be compared against the low-data region of that work's own scaling curve rather than against its headline figure.

The closest prior work is segmentation-specific pretraining (Minixhofer et al., 2023; Frohmann et al., 2024); we evaluate its strongest representative against our Arabic-specific members in Appendix A. We participated in all four subtasks under both tracks. The open-track members we uploaded during the evaluation phase were trained on the released development split. The organisers subsequently ruled that the development and test splits may not be used for training in either track. With their agreement we withdrew those entries and resubmitted, after the phase closed, open-track systems whose members are trained on external public data and the gold `train` split alone, combined by a plain average with nothing fitted on any AraSeg split (Section 4). The organisers scored them on the Blind Test set, and Table 1 reports those scores. Our closed-track entries are as uploaded during the evaluation phase, and are ranked on two of the four subtasks.

Second, three label invariants hold on 100% of documents in every variant of every released split; the final token of a document is always a boundary, and wherever the paragraph marker `\n` occurs it is never a boundary while the token preceding it always is. In the paragraph-aware variants these alone recover $\sim$34% of boundaries at perfect precision.

3 System Overview

Tagger. Each ensemble member is a pretrained encoder with a single-logit head over the final subtoken of each word, trained with binary cross-entropy. Windows packed against an exact 512-subtoken budget cover each document. Each window contributes predictions only from a core region, with 64 words dropped at any edge that is not a document edge, so cores tile each document exactly once. We prepend a condition tag (`[PA]`, `[NP]`, `[NoPnx-PA]`, `[NoPnx-NP]`) to every window, letting one model serve all four subtasks and quadrupling the training signal from the same documents.

Encoder. Replacing the baseline `CAMeL-Lab/bert-base-arabic-camelbert-msa` (Inoue et al., 2021) with `aubmindlab/bert-base-arabertv02` (Antoun et al., 2020) was worth $+1.82$ F1, our largest single change. Multilingual encoders

were not competitive, with XLM-RoBERTa-base (Conneau et al., 2020) at -2.83.

Constrained decoding. We enforce the invariants above in the decoder. Their measured contribution once a model is trained is small ($+0.02$ to $+0.18$), since the tagger learns them unaided, but they cost nothing and cannot introduce a false positive.

Combination. Members are combined by a logistic regression over member logits. Word i is a boundary iff $\sigma(b + \sum_m w_m z_{im}) \geq \tau$, followed by the constraints above, where σ is the logistic function, w_m and b are the fitted weights and bias, and $z_{im} = \log \frac{p_{im}}{1-p_{im}}$ is the logit of member m's boundary probability p_{im} at word i. The combiner is fitted on held-out labelled documents, using only the positions the invariants leave free, since the constrained ones are decided before the combiner is consulted. The threshold τ is swept on the same held-out data. The open-track system, resubmitted after the ruling, uses the plainest rule instead: word i is a boundary iff $\frac{1}{M} \sum_m p_{im} \geq 0.5$, followed by the same constraints, with no weight and no threshold fitted on any split.

4 Experimental Setup

Data splits. In the closed track, models are trained on `train` only (174 documents), all four variants jointly. The combiner and threshold are fitted on `dev`; the public `test` split is reserved for evaluation. No external corpora were used in the closed track.

Open-track data. Silver sentence boundaries are derived from terminal punctuation in Arabic Wikipedia, then the four AraSeg variants are produced by the corpus's own two transforms, dropping punctuation tokens and dropping the paragraph token, so a boundary always sits on the last surviving token of its sentence. Tokenisation matches AraSeg: whitespace split, then every punctuation character peeled off as its own token. Each open-track member is pretrained on 3.2M silver tokens and then fine-tuned on the gold `train` split alone. No AraSeg `dev` or `test` document enters training, and the open-track ensemble fits nothing on either split.

Preprocessing. Words are tokenised independently and pooled at their final subtoken. Two tokens encode to *zero* subtokens under standard BERT tokenisers and would silently shift every subsequent label. The paragraph marker `\n` is remapped to a dedicated added token, and `U+FFFD`, which occurs five times in one test document and never in training or development, is mapped to `[UNK]`.

Metric. The official metric is boundary F1, where a word is positive if it ends a sentence, macro-averaged over documents. We report it throughout and verified our scorer against the organisers' released evaluation script. Significance is a paired Wilcoxon signed-rank test over the per-document scores of the held-out folds.

Hyperparameters. AdamW, learning rate 3×10^{-5}, weight decay 0.01, linear one-cycle schedule with 10% warmup, 5 epochs, effective batch size 16 via gradient accumulation, fp16, gradient clipping at 1.0, maximum sequence length 512, window margin 64. Combination is ℓ_2-regularised logistic regression (C=1). Thresholds are swept on a 0.05 grid. All experiments ran on a single NVIDIA RTX 3080 (10 GB). Unless stated otherwise, each configuration is a single run.

Ensembles. The closed-track system is a 21-member ensemble spanning seven pretrained encoders, varied by seed and pooling scheme (Appendix D lists them). The ensemble grew from 10 members to 16 and then 21; the weights below are from the 10-member fit. The encoders are AraBERTv02 base and large, CAMeLBERT-MSA, ARBERTv2 (Abdul-Mageed et al., 2021), AraELECTRA (Antoun et al., 2021), `asafaya/bert-base-arabic` (Safaya et al., 2020), and XLM-RoBERTa-base. The open-track system is a three-member ensemble of AraBERTv02, AraELECTRA and ARBERTv2, each silver-pretrained as above, combined by the uniform average at 0.5.

5 Results

Table 1 gives the official Blind Test set scores and ranks. One thing is worth stating plainly. Our two ranked closed-track placings are fifth and seventh, and our open-track scores are at or below our closed-track ones on all four subtasks, so nothing here shows combination is what separates the best systems from the rest. The winning systems scored 95.3 on PA and 89.1 on NoPnx-NP, so our closed entries trail them by 0.9 and 2.6 F1 and clear the organisers' baseline (93.9 and 80.7) by 0.5 and 5.8.

	Closed		Open	
Subtask	F1	Rank	F1	Rank
PA	94.4	7 / 12	93.2	6 / 7
NP	91.1	–	90.6	5 / 6
NoPnx-PA	89.9	–	89.7	5 / 6
NoPnx-NP	86.5	5 / 10	86.5	4 / 7

Table 1: Official Blind Test set F1 and rank. Ranks and counts are as printed in the final results published on the shared task website, read on 13 September 2026; the organisers' baseline is ranked among the entries. Every closed-track F1 is the organisers' score for the same `dev`-fitted system, uploaded during the evaluation phase; on NP and NoPnx-PA that submission is not in the final closed-track table, so no rank is available and – is shown.

Subtask	Best single	Avg. (prob)	Avg. (logit)	Learned
PA	94.14	94.35	94.42	94.60
NP	92.74	93.03	92.92	93.01
NoPnx-PA	85.84	87.11	86.91	87.31*
NoPnx-NP	82.80	84.57	84.36	85.06*

Table 2: Combination rules over the same 21 members, 5-fold cross-validated by document over the 222 development documents. Fold standard deviations do not exceed 1.2. * learned combiner beats the probability average at $p<0.05$, paired Wilcoxon over documents. Averaging is worth $+0.89$ mean over the best single member, and learning the weights adds $+0.23$ on top of that. Rows are ordered by increasing member disagreement, as in Table 1.

Table 2 isolates the combination stage. The best single member is included as the floor any combination rule has to clear.

Standalone accuracy does not predict ensemble contribution. Our AraBERTv02-large member scored 0.6 F1 *below* the base model in isolation (fifth-weakest of that ten-member fit), yet carries the largest weight in five of the six weight groups ($+0.24$ to $+0.36$), 1.6 to 3.3 times the runner-up where it leads. Across the full 21-member ensemble, however, combination weight tracks neither standalone F1 (Spearman $+0.17$, $p=0.13$, pooled over variants) nor error correlation (-0.13, n.s.), so coefficients over collinear members are not importance scores and attribution should rest on outcome measurements instead.

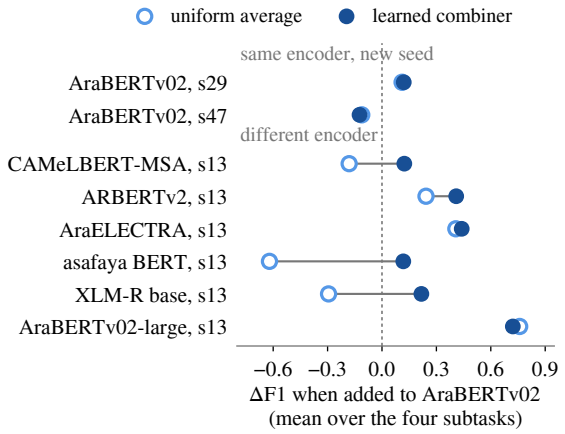


Figure 1: Change in F1 from adding one member to the base encoder, under each combination rule, on the 5-fold development protocol of Table 2. Three of the six different pretrained encoders lower an average and raise a learned combiner; the two extra seeds of the base encoder fall either side of zero.

Every membership decision is small, and so is the choice of rule. Leave-one-out attribution over all 21 members, on the folds of Table 2 (Figure 2), bounds every single membership decision within ± 0.33 F1 under either rule, and under the learned combiner no member costs more than 0.24. Switching rules is of the same order, -0.02 to $+0.49$, so we cannot claim the rule dominates the members it combines. On PA and NoPnx-PA the largest single membership decision is in fact slightly larger than the gain from switching rules. Ensemble size is equally inert, with growth from 10 members to 16 worth nothing to either rule (-0.01 and -0.02). What changes under the combiner is the direction of these decisions rather than their size. The member that most consistently damages an average is not a weak encoder but the smaller of the two synthetic-augmentation models of Appendix A, which ranks worst on three of the four subtasks, echoing the dose-response measured there. On NP the weakest and third-weakest standalone members rank third and first most valuable under the combiner ($+0.21$, $+0.26$) while being merely neutral under averaging (0.00, -0.02), so standalone quality misranks ensemble contribution under one rule and not the other.

Encoder diversity pays, and only a learned combiner collects it. Figure 1 adds each candidate to the base encoder. Under the learned combiner a different pretrained encoder is worth $+0.34$ F1 on average and every one of the six helps ($+0.12$

to $+0.72$); under uniform averaging the same six are worth $+0.05$ on average, and three of them are actively harmful, by as much as 0.62. That is the curation argument in miniature. Averaging obliges you to screen candidates, because half of them cost you accuracy. The combiner makes all six safe to admit. The two extra seeds we tried straddle zero, near ± 0.12 under either rule. The combiner's advantage on this comparison splits by punctuation rather than running as a gradient, -0.02 to $+0.14$ on the punctuated subtasks against $+0.47$ to $+0.56$ on the no-punctuation ones. Practically, this is the one place where the combination rule changes a decision rather than a decimal, removing the admission test.

Error analysis. What makes a boundary hard is not the absence of a punctuation mark on it but the absence of punctuation from the document. In the submitted ensemble's held-out predictions, excluding the constrained positions of Section 3, word-based boundaries in PA reach 93.5 F1, *above* the 90.8 of punctuation-marked ones, yet word-based decisions fall to 81.4 in NoPnx-PA, where every boundary is word-based because the document has no punctuation at all. Word-based segmentation accounts for 39% of sentence boundaries in the corpus (Elkholy et al., 2026a). Mean pairwise member disagreement rises monotonically across the four subtasks (0.65%, 0.88%, 2.02%, and 2.44% of unconstrained positions on `dev` for PA, NP, NoPnx-PA, and NoPnx-NP). This order does not exactly match combination gains at the subtask level: NP has the second-lowest disagreement of the four subtasks yet the only negative gain (-0.02 F1 versus probability averaging), so subtask-level disagreement alone is not the mechanism. It is visible instead within a subtask. Across the 222 held-out documents of each variant, documents where members disagree more are documents where the combiner gains more, in the two no-punctuation subtasks (Spearman $\rho = +0.27$, $p<0.001$ for NoPnx-PA; $\rho = +0.18$, $p=0.009$ for NoPnx-NP) but not detectably in the punctuated ones ($\rho = +0.10$, $p=0.14$ for PA; $\rho = +0.07$, $p=0.32$ for NP). These are four tests, and the two significant ones survive Benjamini-Hochberg at $q = 0.05$. Seed variance does not separate the groups, with three seeds of an identical configuration differing by 0.06 and 0.31 F1 on PA and NP against 0.46 and 0.27 without punctuation. Punctuation, where present, dominates the decision and models agree, leaving a combiner nothing to exploit. In its absence, models rely on lexical cues, disagree in different places, and combination recovers real signal.

Open track: silver pretraining, and the same findings again. All figures here tune the single-model threshold on `dev` and score the held-out public `test` split; the submitted ensemble uses 0.5. A train-only AraBERTv02 on the current library stack averages 88.44. Pretraining it on 3.2M silver tokens and then fine-tuning on gold `train` alone reaches 89.02 ($+0.58$), the gain concentrated on the no-punctuation subtasks ($+0.86$, $+0.94$) where silver boundaries survive the punctuation strip and supply supervision exactly where the corpus is data-starved. The recipe matters more than the volume (Appendix C). Mixing the same silver into the gold set instead peaks at $+0.41$ with 1M tokens and decays to $+0.15$ at 3.2M as Wikipedia's conventions take over, and the two-stage recipe itself decays from $+0.58$ at 3.2M to $+0.30$ at 12M, so silver has an optimum of a few million tokens. Two closed-track findings reproduce on this external data. The two added encoders are weaker standalone than AraBERTv02 (88.40 and 88.61) yet the three-member average clears the best single member by 0.60 (89.62), so standalone accuracy again fails to predict ensemble contribution. ARBERTv2 reaches a training loss of 0.0022 against AraBERTv02's 0.0060 while scoring 0.41 lower on test, overfitting to 174 documents rather than modelling them better. A learned combiner over the same three members, fitted on `dev`, scores 89.68 against the average's 89.62, but the resubmitted system fits nothing on any AraSeg split, so the average was submitted. On the Blind Test set the average scores 89.97.

6 Conclusion

A conventional tagger placed fifth and seventh on the two closed-track subtasks where our entry is ranked, and the accounting is short. Encoder choice supplied $+1.82$ F1, averaging $+0.89$, learned weights $+0.23$ more, and in the open track, silver pretraining $+0.58$, with training-set size dominating all of them. Inference-time structure gave nothing; augmentation cost us. Future systems should spend their effort on data and encoder first, then learn the combination weights rather than average them, so heterogeneous members need no screening.

Limitations

Most development figures were tuned and evaluated on the same documents and are therefore optimistic. Only Table 2 and Figures 1–2 are cross-validated. Most comparisons use a single random seed. Our closed-track combination weights are fitted on `dev`, so no closed-track number here rests on the training split alone. The open-track systems fit nothing on any AraSeg split, but their single-model figures use thresholds tuned on `dev`, and their scores on the Blind Test set were produced by the organisers after the evaluation phase closed, then placed in the final tables alongside entries scored during it.

Four further caveats bear on the central claim. The 21-member ensemble was itself assembled during development with `dev` visible, so while the cross-validation in Table 2 is honest about the combiner, it cannot be honest about the choice of members. Every combination result is a development measurement, since the Blind Test set was scored only for whole submitted systems, and against the transfer offsets of Appendix B a $+0.23$ mean gain is too small to claim it survives at all. A difference smaller than the seed spread of Section 5, 0.06 to 0.46 F1 across the four subtasks, is not distinguishable from the noise of a single training run. And all of this is one corpus in one language, and we have no evidence that the ordering of these levers holds elsewhere.

Ethical Considerations

This work is a preprocessing study on the organisers' released AraSeg corpus; it involves no new data collection, no human annotation, and no personal data beyond whatever the corpus itself contains. All models are publicly available pretrained encoders fine-tuned on that corpus, and in the open track on silver boundaries derived from public Arabic Wikipedia text, on modest compute (a single consumer GPU). Sentence segmentation output carries little direct misuse potential, but our systems inherit the corpus's genre mix and the predominantly MSA pretraining of their encoders. Their behaviour on dialectal Arabic and out-of-domain text is uncharacterised, and readers should not assume the reported accuracy transfers there.

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A Negative Results

Inference-time structure is exhausted. Three independent interventions were null. Per-document expected F1 decoding cost 0.05, minimum-segment-length constraints were negative on every variant, and an exact semi-Markov CRF (Sarawagi and Cohen, 2004) gained 0.06.

Why the semi-Markov CRF fails. We scored a segmentation as the log-likelihood under the model's marginals plus $\lambda \sum \log P(L)$ over segment lengths, with exact dynamic programming over segment endpoints. At $\lambda{=}0$ this reduces to thresholding, so the prior is the only addition. Three follow-up measurements locate the failure.

No entropy at which a corpus-level prior helps. The empirical length distribution has entropy 3.22–3.32 nats across variants (effective support 25–28 lengths, modal probability 0.13), so $\log P(L)$ varies by $\sim$2.6 nats where the tagger's logits exceed ± 5. Temperature-sharpening the prior to force informativeness is monotonically harmful on all four subtasks, costing 0.47–1.33 F1 at moderate sharpening ($T{=}0.1$) and 1.7–3.4 as the prior approaches a point mass. The distribution is right-skewed (median 7–9 against a mode of 4 on the punctuated variants and 1 on the others), so concentrating mass at the mode drives over-segmentation that the marginals resist. The two paragraph-aware variants, where the invariants of Section 3 pin a third of all boundaries, suffer roughly half the damage of the unconstrained variants at equal prior sharpness, so the verified constraints also shield the decoder from a mis-specified soft prior.

The information exists, but is document-specific. An oracle prior built from each document's own gold sentence lengths gains $+2.4$ to $+3.9$ F1 ($+3.1$ mean), more than any single intervention in this work. A realistic middle case that knows only each document's *mean* sentence length (the corpus shape rescaled to match it) recovers just 4% of that headroom.

Much of the oracle is the count. Decoding under a hard constraint to the gold *number* of sentences, with a completely flat length prior, recovers 25–57% of the oracle gain (mean 38%). The remainder requires the length distribution proper. Since neither the exact count nor the length multiset is separable from the segmentation itself, the oracle gain is a measured ceiling on segmental decoding rather than an attainable method.

Synthetic augmentation harms, with a dose-response. Resampling sentences from `train` into synthetic documents cost 0.53 F1 at $\sim 1.7\times$ data and 0.70 F1 at $\sim 2.8\times$, indicating the closed-track ceiling is an information limit rather than a supervision shortage.

Segmentation-specific pretraining does not overcome a monolingual gap. Segment any Text (SaT; Frohmann et al., 2024) is pretrained to predict naturally occurring newlines. Fine-tuned on the training split alone, under the protocol of the encoder comparison above, it averaged 87.85 on development (87.52 on test at the development-tuned thresholds), far above the 61.3 reported for it zero-shot by Elkholy et al. (2026a) but 1.16 below AraBERTv02-base, with the deficit widening from -0.14 on PA to -2.07 on NoPnx-NP. We attribute this to its XLM-RoBERTa backbone. We further note that two of its three robustness mechanisms, inherited from Where's the Point (WtP; Minixhofer et al., 2023), correspond to the organisers' own $\text{CAMeLBERT}_{\text{PnxDrop}}$ and $\text{CAMeLBERT}_{\text{PnxInsert}}$ variants, which they report give no gains over their base model.

B Transfer to the Blind Test set

Each Blind Test set score costs a submission from a limited budget, so the observations here are few and we report them as working practice rather than as an established result.

Scoring the `dev`-fitted combiner of Table 1 on the held-out public `test` split lets us pair a local score with the corresponding Blind Test set score

Recipe	Silver	PA	NP	NoPnx-PA	NoPnx-NP	Mean
none	–	93.64	91.94	85.26	82.92	88.44
mixed	1M	93.84	92.00	85.81	83.75	88.85
mixed	3.2M	93.40	91.79	85.77	83.41	88.59
two-stage	3.2M	93.83	92.28	86.12	83.86	89.02
two-stage	12M	93.68	91.68	85.90	83.69	88.74

Table 3: Public test F1 of a single AraBERTv02 member by silver recipe and volume, threshold tuned on `dev`. *Mixed* adds the silver documents to the gold training set; *two-stage* pretrains on silver and then fine-tunes on gold `train` alone. The two-stage 3.2M row is the recipe of the submitted open-track members.

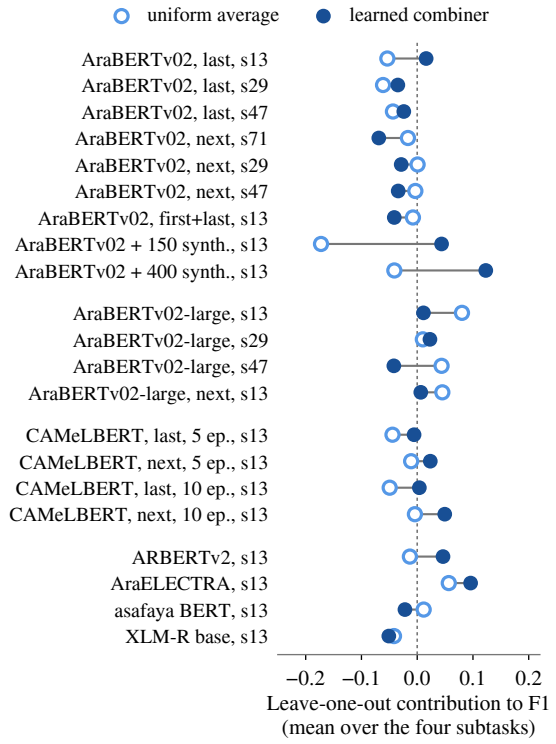


Figure 2: Leave-one-out contribution of each member, the ensemble's F1 minus its F1 without that member, under each combination rule. Rows follow Table 4, and each value is a mean over the four subtasks. Every value lies within ± 0.2; the one member that costs an average more than a tenth of a point is the smaller synthetic-augmentation model, and the combiner turns it positive.

per subtask, on all four. The offsets are $+0.30$ (PA), -1.35 (NP), $+2.54$ (NoPnx-PA) and $+1.08$ (NoPnx-NP): a spread of nearly 4 F1, larger than every effect measured in this paper, and not of one sign. A local score therefore predicts a Blind Test set score only to within a point or two, and only per subtask.

We treated differences below 0.5 F1 as unmeasured when choosing what to submit, and suggest others do the same until someone measures the transfer properly.

The resubmitted open-track system of Table 1 gives a second set of paired observations, with nothing fitted on either side. Its public-test scores are 93.85, 92.52, 87.16 and 84.94, against 93.17, 90.59, 89.66 and 86.47 on the Blind Test set, so the offsets are -0.68, -1.93, $+2.50$ and $+1.53$. The sign agrees with the closed-track offsets on three of the four subtasks, and the spread is again over 4 F1.

C Silver Pretraining Volume

Table 3 gives the open-track silver experiments behind Section 5: one AraBERTv02 member, seed 13, with silver Wikipedia boundaries either mixed into the gold training set or used for a pretraining stage before fine-tuning on gold `train` alone. Both recipes peak at a few million tokens and decay beyond it.

D Ensemble Members and Attribution

Table 4 lists the 21 closed-track members with their standalone development scores, and Figure 2 gives each member's leave-one-out contribution under both combination rules, on the folds of Table 2. All members are trained on `train` alone with the hyperparameters of Section 4 unless the table says otherwise.

Encoder	Pooling	Seed	Notes	Dev F1
AraBERTv02-base	last	13		89.02
AraBERTv02-base	last	29		88.97
AraBERTv02-base	last	47		88.82
AraBERTv02-base	next	71		89.09
AraBERTv02-base	next	29		89.14
AraBERTv02-base	next	47		89.02
AraBERTv02-base	first+last	13		88.77
AraBERTv02-base	last	13	+150 synthetic docs/variant	88.49
AraBERTv02-base	last	13	+400 synthetic docs/variant	88.31
AraBERTv02-large	last	13		88.41
AraBERTv02-large	last	29		88.79
AraBERTv02-large	last	47		88.17
AraBERTv02-large	next	13		89.10
CAMeLBERT-MSA	last	13		87.20
CAMeLBERT-MSA	next	13		87.47
CAMeLBERT-MSA	last	13	10 epochs	87.00
CAMeLBERT-MSA	next	13	10 epochs	87.31
ARBERTv2	last	13		88.21
AraELECTRA	last	13		88.57
asafaya BERT	last	13		85.90
XLM-R base	last	13		86.19

Table 4: The 21 closed-track ensemble members, all trained on `train` for 5 epochs unless noted. Pooling: *last* uses the final subtoken of each word, *next* concatenates it with the first subtoken of the following word, *first+last* concatenates the first and final subtokens. Synthetic documents are resampled from `train` sentences (Appendix A). Dev F1 is the member's standalone score, mean over the four subtasks, with its threshold tuned on `dev`.